PRISM

An AI assistant for in-line spectroscopy solution design:
from a customer enquiry to a grounded first technical assessment.

Deterministic spectroscopy rules · AFS evidence cards with verbatim quotes · clarifying questions · verified claims · feasibility plan

APPROACH

A sales engineer's first assessment, but grounded and honest

The task

- Read a free-text customer enquiry (process, sample, targets, constraints).
- Ask what is missing, recommend method(s), spectral range and probe concept.
- Justify with the AFS materials, state risks and plan a first feasibility study.

What can go wrong with a plain LLM

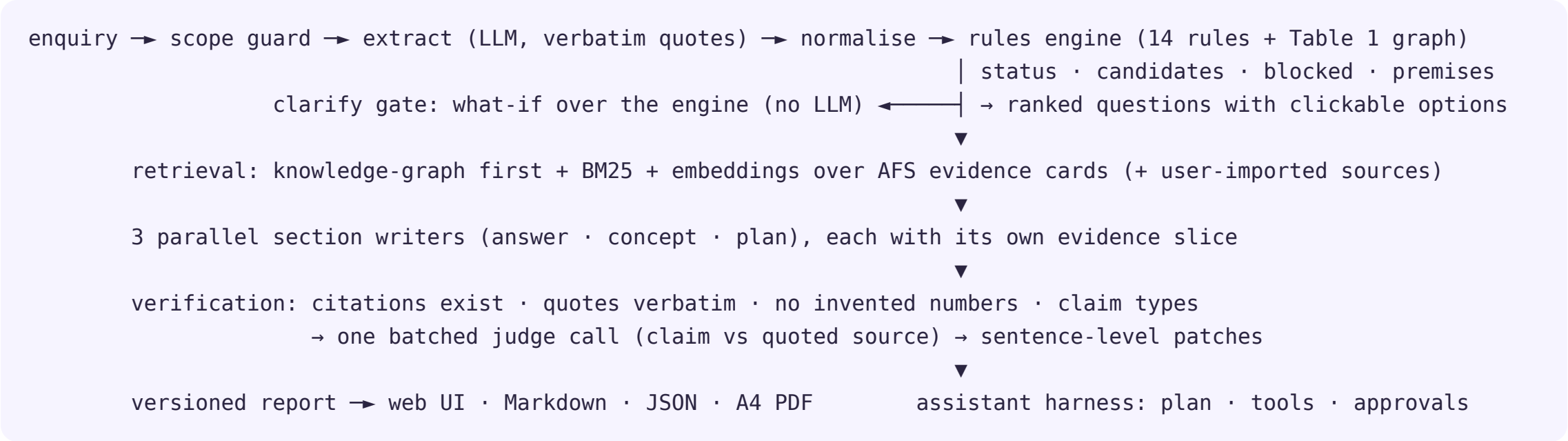
- Invented specifications, ranges or performance figures.
- Missing client facts filled in as if given; lab studies quoted as proof.
- Confident answers where "not enough information" is correct.

Our design principles

- **Rules decide, the LLM writes.** A deterministic engine owns status, candidates and blocked methods.
- **AFS materials first.** Every AFS claim cites a card whose quote is verbatim on the cited page.
- **Typed statements.** Client fact · AFS · external · assumption · proposal · unknown.
- **Ask before guessing.** Questions ranked by how much the answer changes the recommendation.
- **Verify every sentence** before the scientist sees it.

SYSTEM ARCHITECTURE

Neuro-symbolic pipeline with a verification cascade



Decision statuses: PROVISIONAL INSUFFICIENT INFORMATION OUTSIDE SUPPORTED SCOPE , set by the rules; the LLM can only make them stricter.

Provider-agnostic: roles (extractor, writer, judge, agent…) map to models per profile: OpenAI (used for the answers), Gemini, GWDG or a custom endpoint. Every call is cached for replay.

USE OF AFS MATERIALS

The AFS materials are the knowledge base

From materials/ to kb/ (make ingest)

- **Pages:** S01–S05 stored as text and images; citations open on the highlighted page. S02 slides transcribed with a vision model.
- **Method table** (S01 Table 1, xlsx) → technique graph: geometry, sample types, optical path, strengths.
- **71 curated evidence cards:** verbatim quote, page, scope (process application, lab study, in-vitro PoC, general). Quotes are checked against the page at load.
- **14 rules** (sample geometry, optical state, fiber windows, aqueous media, fluorescence targets, calibration transfer, evidence scope, link length…), each citing its cards.

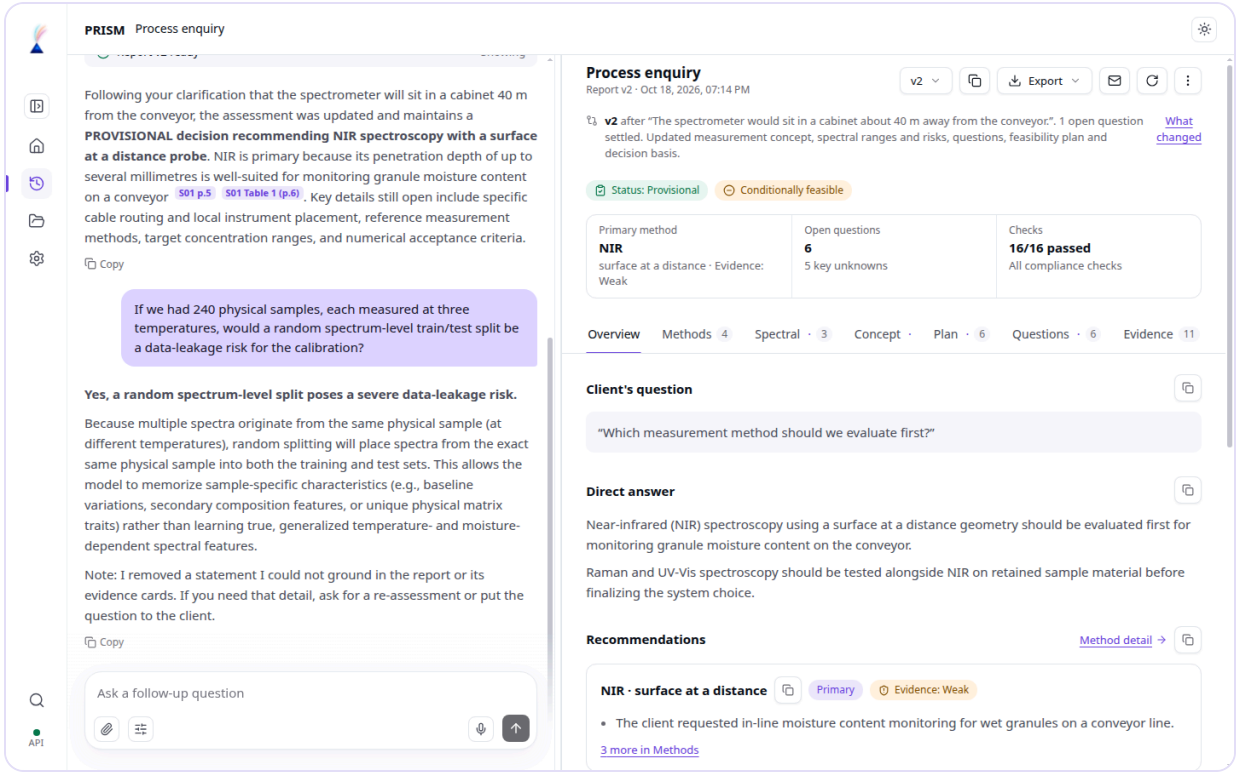
How answers use them

- Rules read the enquiry facts and the graph: e.g. silica fiber 0.35–2.4 µm vs MIR 2.5–25 µm → *contradicted*.
- Writers only see the cards retrieved for their section; citations like **S01 p.10**, **S01 Table 1 (p.6)**.
- External sources only on request (web search, MCP) and always with the quoted passage, marked *external*, never AFS.

No case-specific hard-coding. H01–H06 are read only by the answer script; an n-gram check confirms no prompt, rule or ontology shares wording with them.

THE PROTOTYPE

An assistant for scientists, not a static report



- **Clarifying questions** with clickable options; each option shows what it would change.
- **Living report:** a new fact in the chat revises only the affected sections → Report v2 with "What changed"; major changes ask first.
- **Grounded follow-ups:** distinguishes AFS evidence, client facts, deductions and general knowledge instead of refusing.
- **Exports:** A4 PDF, Markdown, JSON of the version on screen.
- **Your sources:** upload files and folders or add links; cited separately.
- **MCP integrations** with per-tool permissions and approvals.

EXAMPLE RECOMMENDATION · H04

"Can the assistant approve this concept?" Not yet.

Enquiry: water content in a petroleum line using MIR ATR; analyzer in another building; reuse an existing silica fiber from a VIS/NIR setup; link length and conditions not supplied.

Assessment

INSUFFICIENT INFORMATION NOT YET

> There is not enough information to approve the MIR ATR concept as proposed.

Premise check

CONTRADICTED reuse of the silica fiber:

> VIS–NIR silica transmits about 350–2400 nm, which does not overlap the MIR range of 4000–400 cm^{-1} (2.5–25 µm).

Concept

- MIR ATR probe (conditional) and a screening of candidate methods on retained samples.
- Fiber link: polycrystalline (AgHal) IR fiber or hollow glass waveguide; long link to be confirmed with AFS.

First questions to the client

- What does the petroleum liquid look like at the measurement point?
- Cable route and length to the analyzer in the other building?
- Research, pilot or production use?

Two dairy streams, two probe geometries, one comparison study

Recommendation

- **Clear whey stream:** NIR transmission / transflection probe (primary).
- **Opaque milk concentrate:** NIR diffuse reflection probe (primary).
- Spectral range NIR 780–2500 nm; VIS-NIR silica fiber; link length to be confirmed.

> Compare the specified NIR probe geometries on the same representative samples from each stream before selecting the installation.

First feasibility study (≤ 6 steps)

1. Clarify ranges, acceptance criteria and installation constraints.
2. Collect representative samples across agreed compositions and physical states.
3. Obtain time-aligned laboratory references for relevant variables.
4. Set up interfaces and compare spectra and models within each stream.
5. Validate using independent production material.
6. Decide go or no-go against pre-agreed criteria.

FINAL EVALUATION SCENARIOS

H01–H06: generated by PRISM, reproducible by replay

CASE	TOPIC	STATUS	ANSWER	KEY POINT
H01	New formulation after an NIR pilot	PROVISIONAL	Conditional	Test the old calibration on the new blend first; that adding Raman keeps it valid is unverified
H02	Two dairy streams	PROVISIONAL	Conditional	Transmission for whey, reflection for concentrate; compare on both
H03	Fluorescence for glucose and biomass	PROVISIONAL	Conditional	Fluorescence not recommended without a target; NIR / Raman / UV-Vis to compare
H04	Silica link for MIR ATR	INSUFFICIENT	Not yet	Silica fiber reuse contradicted; not approved
H05	Undefined off-spec liquid	INSUFFICIENT	Not yet	No exact configuration yet; screening experiment and questions first
H06	Research sensor for surgery	OUTSIDE SCOPE	No / n.a.	In-vitro study is not proof of clinical readiness; research-grade feasibility only

All six pass every structural check (citations exist, quotes verbatim, no invented numbers, 3–6 questions, plan ≤ 6 steps). **make verify-outputs** replays them from the committed LLM cache and matches byte for byte.

EVIDENCE IT WORKS

Evaluation on development cases and new follow-up questions

7/8
decision status as expected (D01–D08)

0
invented numbers

100%
citations exist, quotes verbatim, client facts kept

~20 s
per report · verdict preview in ~2 s

Gold agreement (D01–D08, live, OpenAI)

- Method family 6/7 · premise verdicts 6/7 · must-cite pages 81 %.
- Judge on every sourced claim: 45 % fully supported, 45 % partly, 10 % unsupported → partly/unsupported sentences are hedged or rewritten before the user sees them.

Generality: 12 new follow-ups on a new domain

- Explanations, unit conversion (1650 cm^{-1} = 6061 nm → IR fiber), a hypothetical, a new fact (→ report v2), data-split advice, XRF (partly outside sources), a contradicted premise, a comparison table, an e-mail draft.
- Prompt injection and off-topic requests refused.

LIMITATIONS AND NEXT STEPS

What PRISM does not do yet

Limitations

- **Knowledge coverage:** 71 curated cards; 129 drafted cards await expert review; S05/S06 are business context only.
- **Support strength:** under half of the judged claims are fully supported by their quote; the rest are marked or hedged.
- **Reading the enquiry** can still misplace a fact (2 of 8 development cases).
- No measured spectra, no OCR, single-user prototype; enquiry text is sent to the chosen LLM provider.

Next steps

- Expert review of the drafted cards and page-level AFS retrieval.
- Calibrate the judge against expert labels (tooling in place: make claims / agreement).
- Attach measured spectra from the AFS lab to the feasibility plan.
- Authentication, roles and an on-premise model for NDA enquiries.